
Question Number. 1

Consider a set of n linearly independent vectors $\{\vec{w}_1, \dots, \vec{w}_n\} \in \mathbb{R}^n$. A vector $\vec{u} \in \mathbb{R}^n$ will:

- Option 1. Always be a linear combination of $\{\vec{w}_1, \dots, \vec{w}_n\}$
- Option 2. Never be a linear combination of $\{\vec{w}_1, \dots, \vec{w}_n\}$

Enter the correct option as the numeric number. That is if option 2 is correct then enter 2.

Question Number. 2

Which of the following is/are CORRECT?

u, v , and w are vectors in $\mathbb{R}^n$.

- A. A set $\{u, v, w\}$ is linearly independent if u can not be written as linear combination of v and w .
- B. A set $\{u, v, w\}$ is linearly dependent if u is a linear combination of v and w .
- C. If a set $\{u, v, w\}$ is linearly dependent then u is a linear combination of v and w .
- D. For three vectors $\mathbf{u}, \mathbf{v}, \mathbf{w}$, if $\{\mathbf{u}, \mathbf{v}\}$ is linearly independent, and $\{\mathbf{v}, \mathbf{w}\}$ is linearly independent; then $\{\mathbf{u}, \mathbf{v}, \mathbf{w}\}$ is linearly independent.

☐ A ☐ B ☐ C ☐ D

Question Number. 3

Let a, b be in $\mathbb{R}$. Consider the three vectors

$$\mathbf{v}_1 = \begin{bmatrix} a \\ 0 \\ 0 \end{bmatrix}, \quad \mathbf{v}_2 = \begin{bmatrix} 0 \\ b \\ 1 \end{bmatrix}, \quad \mathbf{v}_3 = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}.$$

For which values of a and b are $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3$ independent?

- A. $a = 0$ and $b = 1$
 - B. $a \neq 0$ and $b \neq 1$
 - C. $a = 0$ and $b \neq 1$
 - D. $a \neq 0$ and $b = 1$
-

Question Number. 4

Let us consider the following three vectors v_1, v_2 , and v_3 in $\mathbb{R}^3$.

$$v_1 = \begin{pmatrix} 1 & 0 & -2 \end{pmatrix}$$

$$v_2 = \begin{pmatrix} -1 & 0 & 2 \end{pmatrix}$$

$$v_3 = \begin{pmatrix} 1 & 0 & 2 \end{pmatrix}.$$

- A. The three vectors v_1, v_2 , and v_3 are linearly independent.
- B. Every pair of vectors $\{v_1, v_2\}$, $\{v_2, v_3\}$, and $\{v_1, v_3\}$ are linearly independent.
- C. All pairs except the pair $\{v_1, v_2\}$ are linearly independent.
- D. All pairs except the pair $\{v_1, v_3\}$ are linearly independent.

Your Answer:

- ☐ A ☐ B ☐ C ☐ D

Question Number. 5

Given a set S of vectors in $\mathbb{R}^n$, and set $X(X \subset S)$ and $Y(S \subset Y)$, mark all the statements which are always true:

- A. If the vectors of S are Linearly Dependent, then the vectors of X are also Linearly Dependent.
- B. If the vectors of S are Linearly Independent, then the vectors of X are also Linearly Independent.
- C. If the vectors of S are Linearly Dependent, then the vectors of Y are also Linearly Dependent.
- D. If the vectors of S are Linearly Independent, then the vectors of Y are also Linearly Independent.

- ☐ A ☐ B ☐ C ☐ D
-

Question Number. 6

$$a \begin{bmatrix} 1 \\ 2 \\ 3 \\ 4 \\ 5 \end{bmatrix} + b \begin{bmatrix} -1 \\ 2 \\ -3 \\ 4 \\ -5 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

How many number of pairs (a, b) are there, that satisfy the above equation?

- A. 0
- B. 1
- C. Infinite
- D. 2

Question Number. 7

Which of the following set is/are linearly independent?

- A. $\left\{ \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \begin{bmatrix} 2 \\ 1 \end{bmatrix} \right\}$
- B. $\left\{ \begin{bmatrix} 1 \\ -1 \end{bmatrix} \right\}$
- C. $\left\{ \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}, \begin{bmatrix} 2 \\ 3 \\ 4 \end{bmatrix}, \begin{bmatrix} 1 \\ -1 \\ 2 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \right\}$
- D. $\left\{ \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \right\}$

☐ A ☐ B ☐ C ☐ D

Which of the following is(are) sufficient argument(s) to show that the vectors of set S are linearly dependent?

$$S = \left\{ u = \begin{bmatrix} 1 \\ -2 \\ 7 \end{bmatrix}, v = \begin{bmatrix} -7 \\ 14 \\ -49 \end{bmatrix}, w = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \right\}$$

Treat each option independently, correct option independently should be sufficient to infer that vectors are linearly dependent.

A. $0u + 0v + 1w = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$

B. $0u + 0v = w$

C. $0u + 0v + 0w = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$

D. $7u + (-1)v + 1w = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$

Question Number. 9

Given a set of vectors $S(|S| \geq n)$, with all vectors in $\mathbb{R}^n$, which of the following is a necessary and sufficient condition for the vectors of S to be Linearly Dependent?

- A. Exactly n vectors can be represented as a linear combination of other vectors of the set S .
- B. At least n vectors can be represented as a linear combination of other vectors of the set S .
- C. At least one vector u can be represented as a linear combination of any vector(s) of the set S .
- D. At least one vector u can be represented as a linear combination of vectors (other than u) of the set S .

☐ A ☐ B ☐ C ☐ D

Question Number. 10

Let $\{\vec{\mathbf{u}}_1, \vec{\mathbf{u}}_2, \vec{\mathbf{u}}_3\}$ be a linearly independent set in $\mathbb{R}^3$. Then, for a scalar c , what can be concluded about the set $\{c\vec{\mathbf{u}}_1, c\vec{\mathbf{u}}_2, c\vec{\mathbf{u}}_3\}$?

- A. The set $\{c\vec{\mathbf{u}}_1, c\vec{\mathbf{u}}_2, c\vec{\mathbf{u}}_3\}$ is linearly independent for all values of $c \neq 0$
- B. The set $\{c\vec{\mathbf{u}}_1, c\vec{\mathbf{u}}_2, c\vec{\mathbf{u}}_3\}$ is linearly dependent for all values of $c \neq 0$
- C. The set $\{c\vec{\mathbf{u}}_1, c\vec{\mathbf{u}}_2, c\vec{\mathbf{u}}_3\}$ is linearly independent only if $c = 1$
- D. The set $\{c\vec{\mathbf{u}}_1, c\vec{\mathbf{u}}_2, c\vec{\mathbf{u}}_3\}$ is linearly dependent only if $c = 0$

☐ A ☐ B ☐ C ☐ D

Question Number. 11

Which of the following is/are TRUE?

- A. If $v_1, \dots, v_4$ are in $\mathbb{R}^5$ and $\{v_1, v_2, v_3\}$ is linearly dependent, then $\{v_1, v_2, v_3, v_4\}$ is linearly dependent.
- B. If $v_1, \dots, v_4$ are in $\mathbb{R}^5$ and v_1 is not a linear combination of $\{v_2, v_3, v_4\}$, then $\{v_1, v_2, v_3, v_4\}$ is linearly independent.
- C. If $v_1, \dots, v_4$ are linearly independent vectors in $\mathbb{R}^5$, then $\{v_1, v_2, v_3\}$ is also linearly independent.
- D. All of the above.

☐ A ☐ B ☐ C ☐ D

Question Number. 12

What is the definition of linear independence for a set of vectors $\{\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_m\} \subseteq \mathbb{R}^n$? Let $x_1, x_2, \dots, x_m$ be scalars.

- A. $\{\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_m\} \subseteq \mathbb{R}^n$ are linearly independent if the only solution to $x_1\mathbf{u}_1 + x_2\mathbf{u}_2 + \dots + x_m\mathbf{u}_m = \mathbf{0}$ is $x_1 = x_2 = \dots = x_m = 0$.
- B. $\{\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_m\} \subseteq \mathbb{R}^n$ are linearly independent if there exists a non-trivial solution to $x_1\mathbf{u}_1 + x_2\mathbf{u}_2 + \dots + x_m\mathbf{u}_m = \mathbf{0}$.
- C. $\{\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_m\} \subseteq \mathbb{R}^n$ are linearly independent if the vectors are orthogonal to each other.
- D. $\{\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_m\} \subseteq \mathbb{R}^n$ are linearly independent if the vectors are not scalar multiples of each other.

Question Number. 13

- S1 : A set of two vectors in $\mathbb{R}^n$ is Linealy dependent if at least one vector is a multiple of the other.
 - S2 : A set of n vectors in $\mathbb{R}^n$ is Linealy independent if and only if none of the vectors are a multiple of any other vector.
- A. S1 and S2 both are correct
B. S1 is correct and S2 is incorrect
C. S2 is correct and S1 is incorrect
D. S1 and S2 both are incorrect

Question Number. 14

Which of the following is FALSE?

- A. If $v_1, \dots, v_4$ are in $\mathbf{R}^5$ and $\{v_1, v_2, v_3\}$ is linearly dependent then $\{v_1, v_2, v_3, v_4\}$ is linearly dependent.
- B. If $v_1, \dots, v_4$ are in $\mathbf{R}^5$ and v_1 is not a linear combination of $\{v_2, v_3, v_4\}$, then $\{v_1, v_2, v_3, v_4\}$ is linearly independent.
- C. If $v_1, \dots, v_4$ are linearly independent vectors in $\mathbf{R}^5$, then $\{v_1, v_2, v_3\}$ is also linearly independent.
- D. Any set of 6 vectors in $\mathbf{R}^5$ is linearly dependent.

Question Number. 15

Which of the following is(are) true for the following system of linear equations $\mathbf{AX} = \vec{0}$

$$\begin{bmatrix} 2 & 3 & -5 \\ -5 & -1 & 32 \\ 2 & -4 & -26 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

- A. $\mathbf{X} = [-7 \ 3 \ 1]^\top$ is a solution to the equation $\mathbf{AX} = \vec{0}$.
- B. Showing that $\mathbf{X} = [0 \ 0 \ 0]^\top$ is a solution to $\mathbf{AX} = \vec{0}$ is sufficient to conclude that the columns of $\mathbf{A}$ are Linearly Independent.
- C. Showing that $\mathbf{X} = [-7 \ 3 \ 1]^\top$ is a solution to $\mathbf{AX} = \vec{0}$ is sufficient to conclude that the columns of $\mathbf{A}$ are Linearly Independent.
- D. $\mathbf{X} = [0 \ 0 \ 0]^\top$ is a solution to the equation $\mathbf{AX} = \vec{0}$.